

ABEL GARCÍA RUBIO

Senior Backend Engineer | Python | Distributed Systems | Cloud

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Professional Summary

- Senior Backend Engineer with 8+ years of experience. Expert in legacy modernization and complex migrations, specialized in transitioning monolithic architectures to modern, scalable Python/PostgreSQL environments. Proven track record in introducing testing, observability, and automation to legacy codebases without production downtime.
- Strong analytical and mathematical background (MSc in Physics/Math) applied to complex systems and software reliability.

Technical Skills

- **Backend:** Python (Django, FastAPI), Rust, C++, C#, Matlab, R
- **Cloud:** Private Cloud (self-managed), AWS (EC2, S3, Lambda)
- **Methodologies:** SOLID, TDD, CI/CD
- **DevOps:** CI/CD, GitLab CI, Docker, Kubernetes, RabbitMQ, Kafka
- **Security:** OAuth2, OIDC, Keycloak, Cognito, Logto
- **Observability:** Prometheus, OpenTelemetry, AWS CloudWatch

Professional Experience

› BBVA / GFT

Logroño, La Rioja

Nov 2025 - Present

▶ AI Platform / Senior Backend developer

- Developed project scaffolding and developer tooling for internal AI agent platforms.
- Implemented authentication and authorization mechanisms for AI services, ensuring secure integration with enterprise systems.
- Integrated observability using AWS CloudWatch and Prometheus, enabling monitoring, logging and performance analysis of distributed agent workloads.

› Santander Bank S.A. / NovaQuality

Logroño, La Rioja

Feb 2024 - Nov 2025

▶ Senior Python developer

- Architected and deployed a microservices-based backend, leading the progressive migration of a complex monolithic system to a Python/PostgreSQL stack, ensuring data integrity and zero downtime
- Improved scalability and reliability of financial APIs, ensuring high availability on Kubernetes and reducing service downtime.

› Brainy

Zaragoza, Spain

Apr 2023 - Feb 2024

▶ Senior Backend developer

- Led the incremental refactoring of a legacy monolith into microservices, introducing comprehensive test coverage to untested production code to guarantee stability during the transition.
- Designed and implemented CI/CD pipelines to automate build, testing and deployment workflows for microservices.
- Secured distributed services using Keycloak with OAuth2/OIDC, enforcing strong authentication and authorization practices.

› Aragon Photonics Labs

Zaragoza, Spain

Sept 2021 - Apr 2023

▶ Lead Researcher and Developer Engineer / CTO

- Led a team of 5 engineers in the transition from a monolithic architecture to containerized microservices, improving scalability of real-time signal processing applications.
- Cut release cycles from 3 weeks to 1 week through GitLab CI/CD automation.
- Delivered distributed data pipelines handling terabytes of photonic data, enabling real-time analytics.
- ▶ **Researcher and Developer Engineer** Oct 2020 - Sept 2021
 - Applied SOLID and TDD principles in Python projects to refactor legacy code, improving maintainability and test coverage.
 - Built distributed real-time data pipelines to process terabytes of real-time data, enabling advanced analytics and ML integration.
- ▶ **Project Engineer** Feb 2019 - Oct 2020
- ▶ **Efor, Internet and technology** Zaragoza, Spain
- ▶ **Data analyst** Apr 2017 - Nov 2017

Education

- ▶ **Master's in Physics of Complex Systems** Zaragoza, Spain
UNED Sept 2020 - Sept 2021
- ▶ **Master's degree in mathematical modelling, statistics and computing** Zaragoza, Spain
UNIZAR & UPV/EHU Sept 2016 - Sept 2018
- ▶ **Bachelor's degree in Physics** Zaragoza, Spain
UNIZAR Sept 2011 - Sept 2016.

Projects

- ▶ **SecureChat**

React Python Distributed System

 - Developed a distributed real-time chat system demonstrating advanced WebSocket handling and Keycloak-based authentication/authorization: FastAPI (backend), React (frontend)
 - Architected for scalability with multiple backend and frontend instances communicating via messaging queues (RabbitMQ or Redis).
 - Implemented secure WebSocket sessions with role-based access control and token refresh every 5 minutes using Keycloak.
- ▶ **Observability & Agentic Systems Toolkit**

Python OpenTelemetry MCP Best Practices AI Agents

 - Framework for observability across AI agents, MCP servers, and APIs.
 - Integrated OpenTelemetry for granular distributed tracing of agent workflows.
 - Enforced code quality via pre-commit automation and modular design patterns.
- ▶ **Portfolio website** <https://agrubio.dev>

ASTRO Github pages

 - Built a personal portfolio with blog-style project breakdowns using astro.
 - Features examples of backend architecture, performance benchmarks, and code snippets.

Languages

- Spanish (native),
- English (Full professional proficiency)